Question	Answer	Mark	Mark scheme	Additional guidance
21 (a)	$\frac{1}{4}$	M1	for $f(1) = 3 \times 1^2 + 1$ (= 4) **and** a clear intention to find g("4") **or** for $\frac{4}{\left(3\times1^2+1\right)^2}$ **or** for stating gf(x), eg $\frac{4}{\left(3x^2+1\right)^2}$ oe	
		A1	oe	
(b)	$\sqrt[4]{\frac{48}{x-1}}$	M1	for finding fg(x), eg $3\times\left(\frac{4}{x^2}\right)^2+1$ **or** $\frac{48}{x^4}+1$	
		M1	for start of method to find the inverse of fg(x), eg $y-1=3\times\left(\frac{4}{x^2}\right)^2$ **or** $y-1=\frac{48}{x^4}$ **or** $x-1=\frac{48}{y^4}$ **or** $x-1=3\times\left(\frac{4}{y^2}\right)^2$	
		M1	for $y^4=\frac{48}{x-1}$ **or** $x^4=\frac{48}{y-1}$ **or** for a final answer of $\sqrt[4]{\frac{48}{y-1}}$	
		A1	oe	